



TRINETRA

THE THIRD EYE THAT SEES THE DANGER NO SINGLE SENSOR CAN

AI compound-risk intelligence for zero-harm industrial operations

100% compound recall 0% false-positive 7.4 min mean lead

Data present. Unacted upon.

On 13 January 2025, eight workers died
in a coke-oven-battery explosion at the Visakhapatnam Steel Plant.

The gas sensors had data. The permits were logged. SCADA was running.

No layer connected those signals in time.

8

workers killed at Vizag ·
Jan 2025

~3 / day

deaths in India's
registered factories
(DGFASLI)

0

layers connected those
signals in time

The pattern, in a veteran's words

“In more than 30 years in gas and heavy industry, the worst incidents were never one single failure. They were usually three or four small things lining up at the same time, each looking acceptable on its own. Most of the time, nothing actually flagged that combination before it became a serious event.”

NISHAT MULLA · PLANT MANAGER · 30+ YEARS ACROSS GAS & HEAVY INDUSTRY

TRINETRA — OUR RESPONSE

Trinetra is that experience, encoded — it watches every zone 24/7 and flags the combination the moment it forms, so the catch never depends on the right person paying attention at 3 a.m. We enforce safety culture consistently across every shift and handover.

Independent field validation of the problem — a practitioner, in his own words.

Three green lights. One lethal combination.

● RISING FLAMMABLE GAS

still below its alarm

reads NORMAL

● ACTIVE HOT-WORK PERMIT

an ignition source

reads NORMAL

● PERSONNEL PRESENT

inside a confined space

reads NORMAL

A lethal combination no single sensor flags — detected minutes before it becomes critical.

Split reality at T + 8 min

LEGACY SINGLE-SENSOR

● ALL CLEAR

Every gas reads below its setpoint.

threshold detection only

+6

MIN EARLY

TRINETRA COMPOUND AI

● COMPOUND ALERT

COB-1 critical · breach predicted ~36 min
rising CH4 + hot-work permit + 3 personnel

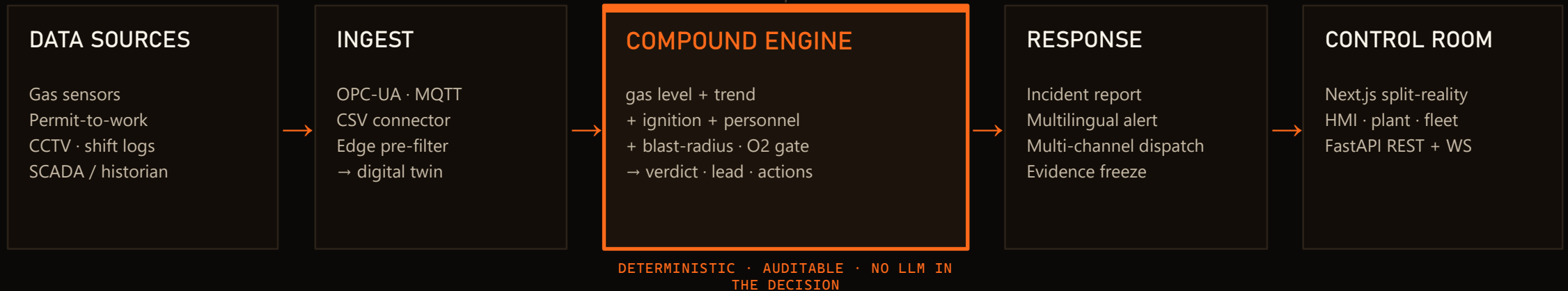
multi-signal fusion

Six minutes of warning — while every gas sensor still reads below its setpoint.

How it works

INTELLIGENCE LAYER · LLM-ASSISTED

Vision (YOLOv8) · Disaster-Memory RAG (Gemini) · 6-stage reasoning graph · Compliance & permit-gate · Pre-mortem · Knowledge graph



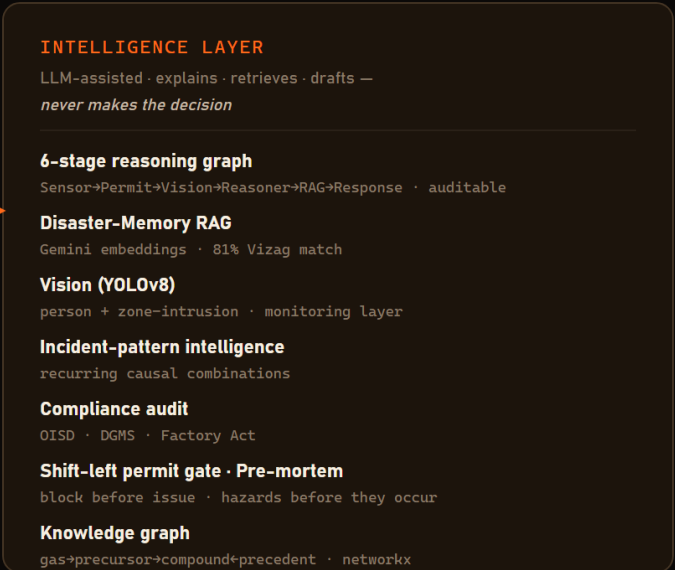
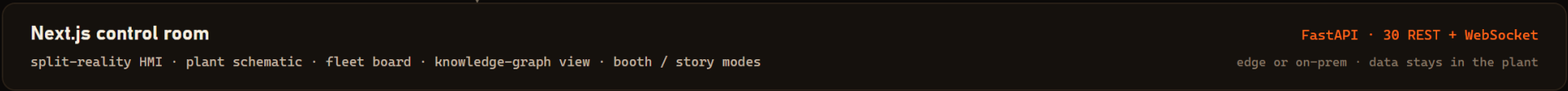
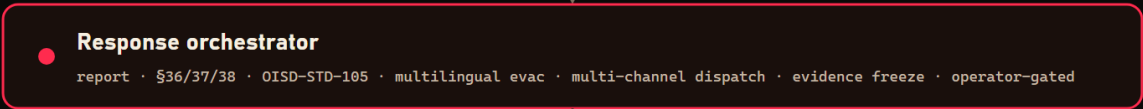
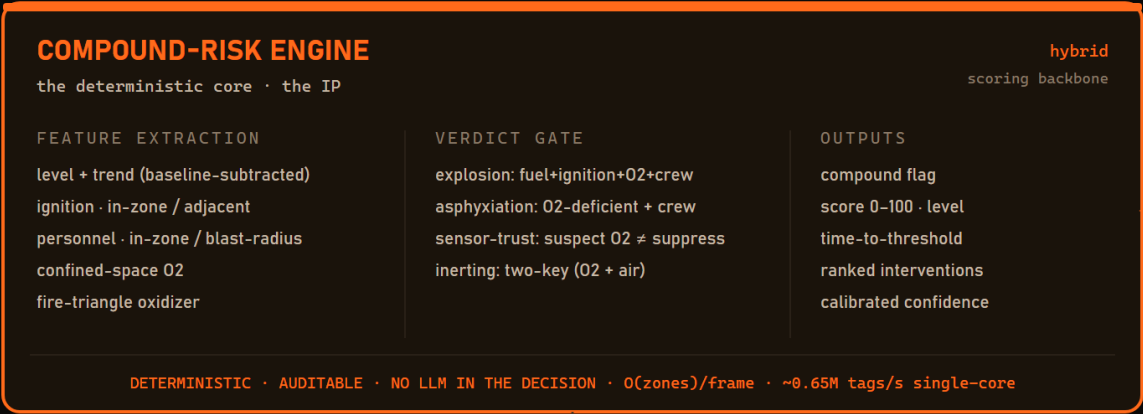
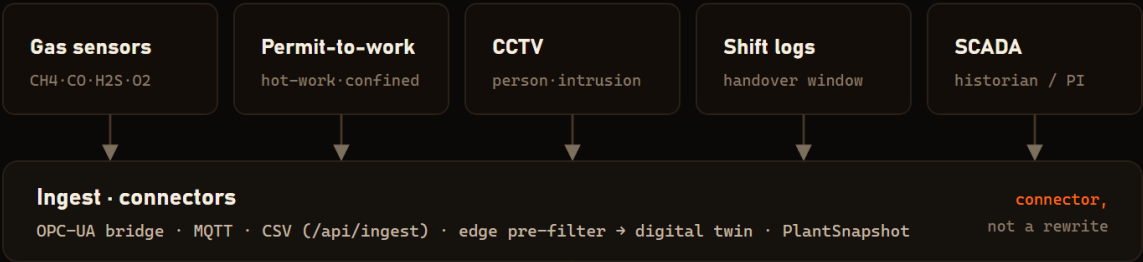
Hybrid by design — the deterministic engine makes every life-safety call; the LLM layer only explains it, retrieves precedent, and drafts reports.



TRINETRA

COMPOUND-RISK INTELLIGENCE · SYSTEM ARCHITECTURE

DATA SOURCES · DIGITAL TWIN



100% compound recall · 0% false-positive · 7.4 min mean lead

29-scenario benchmark · 240-scenario held-out 100% / 3.3% FP · robustness 10/10 · property-tested · deterministic (seed = 42) · vs single-sensor baseline

deterministic decision core

compound-response path

We've seen this death before

81

% MATCH

Visakhapatnam coke-oven explosion · 2025 · 8 killed

GROUNDING BRIEFING

"The present condition echoes the Visakhapatnam coke-oven explosion — rising CH₄ alongside an active ignition source mirrors the precedent's root cause of unacted-upon warnings. The single most important action is the immediate cessation of hot-work and evacuation of personnel."

73%

Texas City refinery

71%

Hot-work ignition

—

Piper Alpha

Live conditions embedded with Gemini, matched against a corpus of real industrial disasters.

The hazard that hasn't happened yet

228

placements searched

136

compound hazards found

124

cross-zone (walkdowns miss)

92

correctly cleared

LARGEST-BLAST-RADIUS DISCOVERY

Rising methane in Coke Oven Battery #1, hot work in the bay, and the crew working next door in the Gas Cleaning Plant — compound CRITICAL by T+10. The blast radius spans the gas-cleaning plant, the confined sump and the maintenance bay. No single zone looks dangerous on its own.

The engine runs as an oracle across the plant's blast-radius map — discovering cross-zone compounds a zone-by-zone walkdown rates safe, before they occur.

Block the permit before it's issued

PERMIT DESK · HOT-WORK REQUEST · COB-1

BLOCKED

HIGH 80



CRITICAL 100 · compound

Issuing this hot-work permit completes the lethal compound pattern — flammable gas + ignition + personnel — minutes before any single sensor would alarm.

CONDITIONS TO ISSUE SAFELY

- Gas-free certificate — flammable gas below alarm and falling (OISD-STD-105)
- Force-ventilate / purge the zone and re-test the atmosphere
- No flammable release developing in this zone or any blast-radius neighbour
- No entry while an ignition source is active in the blast radius (Factory Act §36 / §37)

The inverse of detection: simulate the plant WITH the proposed permit, and refuse it if issuing it would create — or add people to — a compound hazard.

Detect → draft → alert

AUTO-DRAFTED INCIDENT REPORT

- 1 Summary
- 2 Conditions Detected
- 3 Compound Risk Assessment
- 4 Immediate Actions
- 5 Regulatory References
 - Factory Act 1948 §36 / §37 / §38
 - OISD-STD-105 (Work Permit System)
- 6 Corrective Actions

EVACUATION ALERT

Compound hazard in COB-1. Evacuate now. Do not operate hot-work or electrical equipment.

ENGLISH

TELUGU

HINDI

RESPONSE ACTIONS – PREPARED

- ✓ Suspend hot-work permit PTW-7741
- Evacuate & page response team
- Freeze sensor + CCTV evidence
- File preliminary incident report

Measured, not claimed

100%

compound recall

14 / 14 hazards caught

0%

false-positive

0 / 15 incl. inerted + 02 decoys

7.4

min mean lead

median 6 · max 12 · vs baseline

VIZAG RECONSTRUCTION

+6 MIN LEAD



Deterministic (seed=42) · held-out: 100% recall / 3.3% FP over 240 unseen-seed scenarios · property-tested.

— REPLAYED FROM TWO REAL INQUIRIES

Would it catch a real one? Twice.

U.S. CSB · 2005 · Report 2005-04-I-TX

BP TEXAS CITY REFINERY

+10 earlier

min before the documented vapour-cloud ignition (T+20). 15 killed.

MB Lal Committee · 2009

INDIAN OIL JAIPUR DEPOT

+36 earlier

min before the documented ignition (T+48) of a long, undetected vapour build-up. 12 killed.

Same engine, reconstructed inputs. Each inquiry's documented escalation is replayed through the live engine; where a site had no gas detector (a finding in both), the vapour build-up maps to the flammable channel — ignition and personnel are the inquiry's.

The math a buyer underwrites

PER PREVENTED INCIDENT

₹115.5 Cr

avoided loss — a Vizag-class event

Lives protected	₹7.5 Cr
Asset damage	₹40 Cr
Downtime (21 d)	₹63 Cr
Regulatory penalty	₹5 Cr

EXPECTED ANNUAL RETURN

≈ 7.7×

on the ₹1 Cr / plant / yr platform — even at a conservative 1-in-15-year event

3.9×	7.7×	14.4×
1-in-30 yr	1-in-15 yr	1-in-8 yr

+ insurance lever: a recurring ₹0.4–1.2 Cr/yr premium reduction (5–15% for continuous monitoring) offsets the platform cost whether or not an incident ever occurs.

A wedge, a buyer, a trigger

BUYER

Plant process-safety head / safety officer — owns permit-to-work and statutory compliance, carries the fatality risk.

WEDGE

Permit-to-work intelligence — one discrete, fundable pain — then expand to full gas + permit + CCTV + shift fusion.

PRICING

~₹1 Cr / plant / yr — justified against a single downtime-day (~₹3 Cr) or one OISD / NGT penalty.

TRIGGER

Post-LG-Polymers / Vizag regulatory pressure — DGMS scrutiny and mandatory near-miss reporting.

THE ASK

1–2 design partners (steel / refining) for a 90-day pilot on a live permit-to-work + gas feed.

The moat is the fusion, not the sensors

INCUMBENTS SELL SILOS

Honeywell, Dräger, MSA, Hexagon sell gas detection, permit-to-work and CCTV as SEPARATE products. The danger lives in the seam BETWEEN them — exactly what no one owns end to end.

A DETERMINISTIC PATTERN LIBRARY

Compound rules — flammable × ignition × personnel × oxidizer × blast-radius, encoded to real OISD / Factory-Act thresholds — are domain IP, not a model cloned from a public dataset.

A PER-PLANT DATA MOAT

Each plant's operator feedback tunes its own nuisance profile — proprietary data that compounds per site and raises switching cost. An incumbent's fixed hardware doesn't learn.

AUDIT-GRADE BY CONSTRUCTION

A transparent, deterministic core a safety officer can defend in a statutory audit. A pure-LLM bolt-on can't make a life-safety call a regulator will accept.

The wedge: land on permit-to-work intelligence — one fundable pain — then own the fused layer the incumbents are structurally unable to assemble.

— SCALES HORIZONTALLY

One engine. Every plant.

100

plants · one engine

\$0.30

per plant / month

thousands

plants per core

0.05 ms

p50 per assessment

MEASURED \$/PLANT/MONTH — FALLS WITH SCALE

\$3.00

10 plants

>

\$0.30

100 plants

>

\$0.03

1,000 plants

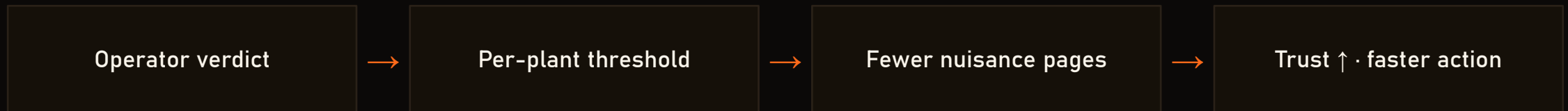
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\$0.009

10,000 plants

Stateless shards. Each plant is independent — no shared state, no per-site model. The engine is O(zones); the fleet scales by adding plain workers. Digital-twin sites stand in for live OPC-UA / MQTT feeds — ingesting real data is a connector, not a rewrite.

Every plant learns its own nuisance profile



RECALL GUARDRAIL

Compound, HIGH and CRITICAL alerts bypass the threshold entirely. Feedback can only ever damp non-compound nuisance pages — never reduce recall.

THE FLYWHEEL

A confirmed false alarm raises the threshold; a confirmed alert relaxes it. Each plant quietly auto-acknowledges its own routine excursions — fewer nuisance pages, faster action on the ones that matter.

The default engine is untouched — the benchmark (100% recall / 0% false-positive) is byte-identical with or without feedback.

Sits over the sensors you already have

Standard formats

Ingests SCADA / IoT / permit-to-work data as-is

A connector, not a rewrite

Digital twin → live plant swaps the data source, not the brain

Auditable & hybrid

Deterministic decisions a safety officer can defend in an audit

WHERE IT DEPLOYS

Steel

coke ovens, blast furnaces

Refining & Petrochem

OISD permit-to-work regimes

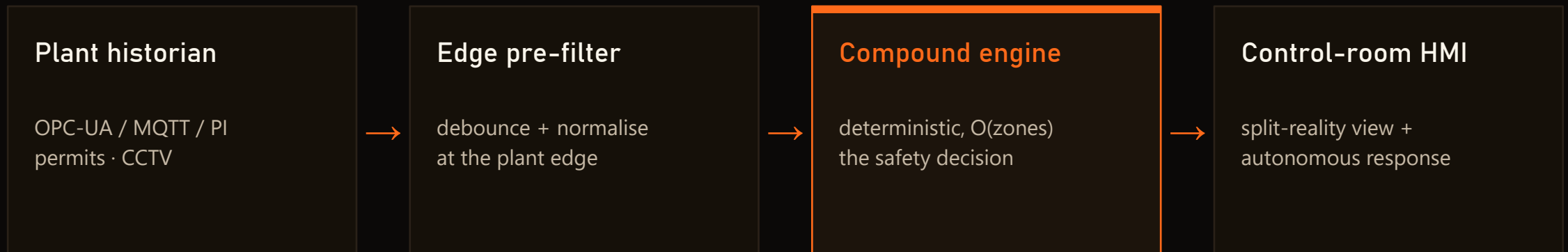
Mining

confined-space & gas hazards

Power & Chemicals

asset-intensive operations

A connector, not a rewrite



THROUGHPUT

~0.65M tags / sec

single-core, measured — a 10,000-tag plant assesses in ~15 ms per 1 Hz frame (~65× real-time). No GPU in the life-safety path.

- **Standard formats**
ingests SCADA / IoT / permit data as-is
- **Edge or on-prem**
data never has to leave the plant
- **Deterministic core**
an auditable decision an officer can defend



Three workers a day.
The data already exists.

Trinetra is the layer that acts — before, not after.

100% recall 0% false-positive 7.4 min lead